Food and Drug Administration
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November 4, 2015

Neusoft Medical Systems Co., Ltd.
℅ Tian Yuehui
Manager of Q&R Department
No.16, Shiji Road, Hunnan Industrial Area
Shenyang, Liaoning 110179
CHINA

Re: K151383
Trade/Device Name: NeuViz 128 Multi-Slice CT Scanner System
Regulation Number: 21 CFR 892.1750
Regulation Name: Computed tomography x-ray system
Regulatory Class: II
Product Code: JAK
Dated: September 17, 2015
Received: September 21, 2015

Dear Tian Yuehui:

We have reviewed your Section 510(k) premarket notification of intent to market the device referenced above and have determined the device is substantially equivalent (for the indications for use stated in the enclosure) to legally marketed predicate devices marketed in interstate commerce prior to May 28, 1976, the enactment date of the Medical Device Amendments, or to devices that have been reclassified in accordance with the provisions of the Federal Food, Drug, and Cosmetic Act (Act) that do not require approval of a premarket approval application (PMA). You may, therefore, market the device, subject to the general controls provisions of the Act. The general controls provisions of the Act include requirements for annual registration, listing of devices, good manufacturing practice, labeling, and prohibitions against misbranding and adulteration. Please note: CDRH does not evaluate information related to contract liability warranties. We remind you, however, that device labeling must be truthful and not misleading.

If your device is classified (see above) into either class II (Special Controls) or class III (PMA), it may be subject to additional controls. Existing major regulations affecting your device can be found in the Code of Federal Regulations, Title 21, Parts 800 to 898. In addition, FDA may publish further announcements concerning your device in the Federal Register.

Please be advised that FDA's issuance of a substantial equivalence determination does not mean that FDA has made a determination that your device complies with other requirements of the Act or any Federal statutes and regulations administered by other Federal agencies. You must comply with all the Act's requirements, including, but not limited to: registration and listing (21 CFR Part 807); labeling (21 CFR Part 801); medical device reporting (reporting of medical device-related adverse events) (21 CFR 803); good manufacturing practice requirements as set forth in the quality systems (QS) regulation (21 CFR Part 820); and if applicable, the electronic product radiation control provisions (Sections 531-542 of the Act); 21 CFR 1000-1050.

If you desire specific advice for your device on our labeling regulation (21 CFR Part 801), please contact the Division of Industry and Consumer Education at its toll-free number (800) 638 2041 or (301) 796-7100 or at its Internet address http://www.fda.gov/MedicalDevices/ResourcesforYou/Industry/default.htm. Also, please note the regulation entitled, "Misbranding by reference to premarket notification" (21 CFR Part 807.97). For questions regarding the reporting of adverse events under the MDR regulation (21 CFR Part 803), please go to http://www.fda.gov/MedicalDevices/Safety/ReportaProblem/default.htm for the CDRH's Office of Surveillance and Biometrics/Division of Postmarket Surveillance.

You may obtain other general information on your responsibilities under the Act from the Division of Industry and Consumer Education at its toll-free number (800) 638-2041 or (301) 796-7100 or at its Internet address http://www.fda.gov/MedicalDevices/ResourcesforYou/Industry/default.htm.

Sincerely yours,

Robert Ochs

Robert Ochs, Ph.D.
Director
Division of Radiological Health
Office of In Vitro Diagnostics
and Radiological Health
Center for Devices and Radiological Health

Enclosure

DEPARTMENT OF HEALTH AND HUMAN SERVICES
Food and Drug Administration

Indications for Use

Form Approved: OMB No. 0910-0120
Expiration Date: January 31, 2017
See PRA Statement below.

510(k) Number *(if known)*
K151383

Device Name
NeuViz 128 Multi-Slice CT Scanner System

Indications for Use *(Describe)*
The NeuViz 128 Multi-Slice CT Scanner System can be used as a whole body computed tomography X-ray system featuring a continuously rotating X-ray tube and detector array. The acquired X-RAY transmission data is reconstructed by computer into cross-sectional images of the body from either the same axial plane taken at different angles or spiral planes taken at different angles.

Type of Use *(Select one or both, as applicable)*

☒ Prescription Use (Part 21 CFR 801 Subpart D) ☐ Over-The-Counter Use (21 CFR 801 Subpart C)

CONTINUE ON A SEPARATE PAGE IF NEEDED.

510(K) Summary

This 510(k) summary of Safety and Effectiveness information is submitted in accordance with the requirements of 21 CFR Part 807.92

General Information:

Trade Name:	NeuViz 128
Common Name:	CT Scanner
CFR Section:	21 CFR 892.1750
Classification Name:	Computed tomography x-ray system
Product Code:	JAK
Classification:	Class II
Performance Standard:	21 CFR Subchapter J, Federal Diagnostic X-ray Equipment Standard
Manufacturer:	Neusoft Medical Systems Co., Ltd. No.16 Shiji Road, Hunnan Industrial Area, Shenyang, Liaoning, 110179, China.
Submitter:	Contact : Tian Yuehui Title : Manager of Q&R Department Tel : 86-24-83660646 Fax : 86-24-83660563 E-Mail : tianyh@neusoft.com

Summary prepared :September 16, 2015

Safety and Effectiveness information

Indications for use:

The NeuViz 128 Multi-Slice CT Scanner System can be used as a whole body computed tomography X-ray system featuring a continuously rotating X-ray tube and detector array. The acquired X-RAY transmission data is reconstructed by computer into cross-sectional images of the body from either the same axial plane taken at different angles or spiral planes taken at different angles.

Device Description:

The NeuViz 128 Multi-slice CT Scanner System is composed of a gantry, a patient couch, an operator console and includes image acquisition hardware and software, and associated accessories. It is designed to be a head and whole body X-ray computed tomography scanner which features a continuously rotating tube-detector system and functions according to the fan beam principle.
The system provides the filter back-projection (FBP) and iterative reconstruction algorithm(ClearView cleared in K133373) to reconstruct images. The end user can choose to apply either ClearView or the FBP to the acquired raw data.
The system software is an interactive program used for X-ray scan control, image reconstruction, and image archive/evaluation. It provides the following digital image processing and visualization tools:

- Support following scan speed:0.374s、0.5s、0.6s、0.8s、1.0s、1.5s、2.0s.
- Surview scan
- Dual surview
- Spiral scan
- Axial scan
- Image reconstruction
- Plan scan
- Patient information management
- Patient information registration
- Protocol selection
- Bolus Tracking
- CCT scan
- Cardiac scan
- Multi-phase scan
- Tube Warm-up
- Film
- Report
- 2D
- DICOM Viewer
- Home
- MPR
- 3D
- Virtual Endoscopy
- Dental
- Vessel Analysis
- Virtual Colonoscopy
- Brain Perfusion
- Body Perfusion
- Lung Nodule Analysis

- Lung Density Analysis
- Coronary Analysis
- Cardiac Calcium Scoring
- Cardiac Function Analysis
- Cardiac Viewer
- Fat Analysis
- CTDSA
- Tumor Assessment

Predicated Device:

Philips Ingenuity Core128 CT System(K033326)

Reference Device

NeuViz 64e CT system (K121792).

Statement of Substantial Equivalence:

The NeuViz 128 Multi-slice CT Scanner System has the same indications for use as the predicate device, “Philips Ingenuity Core128” CT Scanner (K033326). The “NeuViz 128” is a computed Tomography X-Ray System intended to produce images of the body by computer reconstruction of x-ray transmission data taken at different angles and planes. The subject device does not have significant changes in technological characteristics when compared to the predicate device. The subject device and predicate device are the same in regards to:

- Detector/Tube system that can rotate continuously. The Tube emits X-ray beam of fan shape, detector system with multiple channels;
- Low voltage slip ring and On-Board HV Generator;
- Parallel data acquisition, simultaneous scan and image reconstruction;
- Gantry that may be tilted forward or backward;
- Patient Couch that can be elevated or lowered;
- Computerized control of scan and other operations;
- Computerized image reconstruction and post processing;
- Digitarized display, storage and output of data/image.

The subject device and the reference device, “NeuViz 64e” CT Scanner (K121792) are the same in regard to most of application features.

A complete comparison table is included in this submission. The following table shows a brief comparison of the application features, including both similarities and differences among the subject, the predicate and the reference devices.

NeuViz 128	Ingenuity Core128	NeuViz 64e
O-Dose	√	√
Networking	√	√
Bolus tracking	√	√
SAS	√	√

Home	√	√
Filming	√	√
Report	√	√
Image Review	√	√
MPR	√	√
3D	√	√
VE(Virtual Endoscopy)	√	√
Vessel Analysis	√	√
Dicom Viewer	√	√
Barcode Reader	√	√
Dual Monitor	√	√
Continuous CT	√	√
ClearView	√	√
iHD	√	×
Cardiac Scan	√	√
Dental Analysis	√	√
Virtual Colonoscopy	√	√
Brain Perfusion	√	√
Body Perfusion	√	√
Lung Nodule Analysis	√	√
Lung Density	√	√
Coronary Analysis	√	√
Cardiac Calcium Scoring	√	√
Cardiac Function Analysis	√	√
Cardiac Viewer	√	×
Fat Analysis	×	×
Nerve System DSA	×	√
Tumor Assessment	×	√
Preprocessing function	√	×
Comparison		
" √ " symbol means the comparable type and substantially equivalent to the contrasted devices. "×" symbol means the new clinical functionality. But the main algorithm of Fat Analysis Application is segmenting the contour of subcutaneous and visceral, and calculating the area of subcutaneous and visceral according to the result of segmentation. Additionally, the segmentation method is similar to the Contour Segmentation function in 3D Application, and the calculation method is similar to the ROI's area calculation in Common Tool. Verification and validation testing supports this application.		

According to the comparison based on the requirements of 21 CFR 807.87, we stated that these devices are substantially equivalent.

Nonclinical Testing:

The safety and effectiveness of the NeuViz 128 was assured by adherence to Good Manufacturing Practices(GMP) 21CFR 820 and to International Standards ISO 13485:2003. This device is in conformance with the applicable parts of the following standards:

- ANSI/AAMI ES60601-1:2005/(R) 2012 and A1:2012, C1:2009/(R)2012 and A2:2010/(R)2012 medical electrical equipment -- part 1: general requirements for basic safety and essential performance

- IEC 60601-1-2: 2007, Medical electrical equipment -- Part 1-2: General requirements for basic safety and essential performance - Collateral standard: Electromagnetic compatibility - Requirements and tests
- IEC 60601-1-3: 2008, Medical electrical equipment -- Part 1-3: General requirements for basic safety and essential performance - Collateral Standard: Radiation protection in diagnostic X-ray equipment
- IEC 60601-1-6: 2013, Medical electrical equipment - Part 1-6: General requirements for basic safety and essential performance - Collateral standard: Usability
- IEC 60601-2-28: 2010, Medical electrical equipment – Part 2-28：Particular requirements for the basic safety and essential performance of X-ray tube assemblies for medical diagnosis
- IEC 60601-2-44:2009, Medical Electrical Equipment - Part 2-44: Particular Requirements For The Basic Safety And Essential Performance Of X-Ray Equipment For Computed Tomography
- IEC 60825-1:2007, Safety of laser products - part 1: equipment classification, and requirements
- IEC 61223-3-5: 2004, Evaluation and routine testing in medical imaging departments - Part 3-5: Acceptance tests - Imaging performance of computed tomography X-ray equipment
- IEC 62304:2006, Medical device software - Software life-cycle processes
- IEC 62366:2014,Medical Devices - Application Of Usability Engineering To Medical Devices
- ISO 10993-1: 2009, Biological evaluation of medical devices - part 1: evaluation and testing within a risk management process.
- ISO 14971:2007, Medical devices – Application of risk management to medical devices
- NEMA PS 3.1 - 3.20: 2011, Digital Imaging and Communications in Medicine (DICOM)
- NEMA XR 25: 2010, Computed tomography dose check
- NEMA XR 28: 2013, Supplemental Requirements for User Information and System Function Related to Dose in CT

Additionally, this device complies with all applicable requirements of the radiation safety performance standards, as outlined in 21 CFR §1010 and §1020.

Risk management is ensured via a hazard analysis which is used to identify potential hazards. These potential hazards are controlled during product development, and verification and validation testing. To minimize electrical, mechanical, and radiation hazards, Neusoft adheres to recognized and established industry practice and standards.

Software safety is assured by the company procedures that conform to accepted practices. Quality assurance procedures are adhered to, and meeting the specifications and functional requirements if demonstrated via testing. Software Documentation for a Moderate Level of Concern, per the FDA guidance document, “Guidance for the Content of Premarket Submissions for Software Contained in Medical Devices Document” issued on May 11, 2005, is also included as part of this submission.

Verification and validation activities(including performance testing, safety testing and simulated use testing) are performed. The results of these tests demonstrate that the subject device performs as intended. The result of all conducted testing was found acceptable to support the claim of substantial equivalence.

clinical Testing:

Sample clinical images were provided within the submission. The image quality has been evaluated in bind state by two certified radiologists.

Conclusions:

According to the comparison based on the requirements of 21.CFR 807.87, We state that the candidate device and the predicate device are substantially equivalent.